

# A Miyaoka-Yau inequality for hyperplane arrangements in $\mathbb{CP}^d$

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Living on the edge of the moduli space, Aarhus

18-22 May 2026

# Outline

- Introduction: constant curvature metrics on Riemann surfaces with conical singularities

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- Sketch proof: Kobayashi-Hitchin correspondence for parabolic bundles
- Conjectures: Kähler-Einstein metrics and Miyaoka-Yau inequality for log pairs

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- If  $\chi(X, \alpha) > 0$ . Furthermore, assume that  $\alpha_i \in (0, 1)$  and  $n \geq 3$ . Then there exists a unique Kähler metric on  $X = \mathbb{CP}^1$  with  $K_g = 1$  and cone angle  $2\pi\alpha_i$  at  $x_i$  if and only if

$$\forall i \quad a_i < \frac{1}{2} \cdot \sum_{i=1}^n a_i \quad \text{where } a_i = 1 - \alpha_i$$

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## Theorem (dB-Panov, 2025-26)

Let  $\mathcal{H} = \{H_1, \dots, H_n\}$  be a hyperplane arrangement in  $\mathbb{CP}^d$  with  $d \geq 1$ . Let  $\alpha_1, \dots, \alpha_n \in (0, 1)$ . Write  $a_i = 1 - \alpha_i$ . Assume that:

- ①  $\sum_{i=1}^n a_i = d + 1$
- ② the weighted arrangement  $\{(H_i, a_i)\}_{i=1}^n$  is **stable**

Then the following inequality holds:

$$Q(a_1, \dots, a_n) \leq 0 \tag{1}$$

where  $Q$  is the **Hirzebruch quadratic form** of  $\mathcal{H}$ . Moreover, the equality holds if and only if there exists a **polyhedral Kähler (PK)** metric on  $\mathbb{CP}^d$  with cone angles  $2\pi\alpha_i$  along  $H_i$ .

Let  $\sigma_i$  be the number of points  $p$  with  $r_p \geq 3$  lying on the  $i$ -th line of the given arrangement of  $k$  lines in the plane. We consider the  $(k \times k)$ -symmetric matrix  $A$  with

$$(3) \quad A_{ij} = \begin{cases} 3\sigma_i - 4 & (i = j), \\ 2 & (i \neq j, \quad p \in L_i \cap L_j \text{ with } r_p = 2), \\ -1 & (i \neq j, \quad p \in L_i \cap L_j \text{ with } r_p \geq 3). \end{cases}$$

With the  $k$  lines we associate real variables  $x_i$  and let  $x$  be the column vector  $(x_1, \dots, x_k)$ . With the  $s$  points  $p_j$  with  $r_{p_j} \geq 3$  we associate real variables  $y_j$ .

For each point  $p_j$  with  $r_{p_j} \geq 3$  we consider the linear form

$$P_j(x, y) = 2y_j + \sum_{p_j \in L_i} x_i, \quad \text{where } y = (y_1, \dots, y_s).$$

**Höfer's formula.** For the algebraic surface  $Y$  (a good covering of  $S$  of degree  $d$  with respect to  $L_1, \dots, L_k, E_1, \dots, E_s$  and the given branching numbers  $n_1, \dots, n_k, m_1, \dots, m_s$ ) we have

$$(4) \quad (3c_2(Y) - c_1^2(Y))/d = \frac{1}{4} \left( x^t A x + \sum_{j=1}^s P_j(x, y)^2 \right),$$

where  $x_i = 1 - \frac{1}{n_i}$  and  $y_j = -1 - \frac{1}{m_j}$ .



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- $\nabla$  is the Levi-Civita connection of a flat Kähler metric  $g$  on the arrangement complement whose metric completion is polyhedral

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THANK YOU!